10 Stages of the Pipeline:

1. **Problem Definition**
    1. Task: Classify emails as *spam* or *ham (not spam)*.
    2. Performance Measure (P): Classification accuracy.
2. **Data Collection**
    1. Source: Public datasets (e.g., Enron email dataset, SpamAssassin).
    2. Ensure balanced representation of spam and ham.
3. **Data Preprocessing**
    1. Handle missing values (e.g., impute or drop).

    Normalize/scaling numerical features (e.g., email length).

    Text cleaning: remove HTML tags, punctuation, stopwords.

4 Exploratory Data Analysis (EDA)

Visualize spam vs ham distribution.

Word frequency analysis.

Identify correlations between features.

5.Feature Engineering

Extract meaningful features (see feature_list.txt).

Convert text into numerical vectors (TF-IDF, word embeddings).

Create domain-specific features (e.g., presence of suspicious links).

6.Model Selection

Candidate models: Logistic Regression, Naïve Bayes, Random Forest, SVM.

Choose based on interpretability and performance.

7.Model Training

Split dataset into train/validation/test.

8.Model Evaluation

1. Metrics: Accuracy (primary), Precision, Recall, F1-score.
2. Confusion matrix to analyze misclassifications.

9.Deployment

1. Package model into API (Flask/FastAPI).
2. Integrate with email client/server.
3. Ensure scalability and security.

10.Monitoring & Maintenance

- Track accuracy drift over time.
- Retrain with new spam patterns.